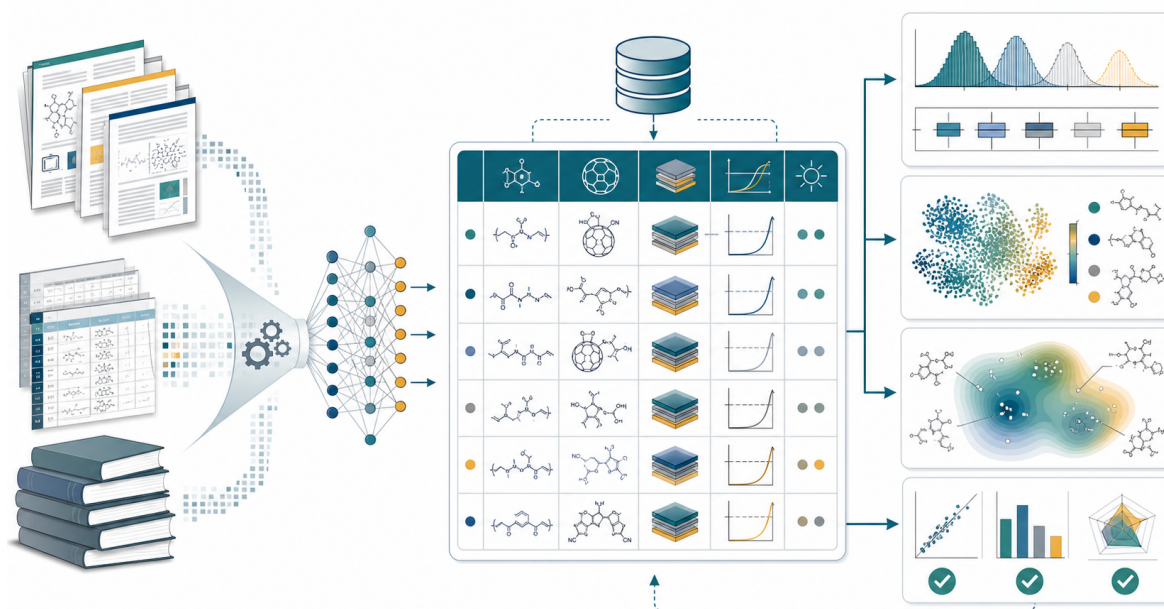


A literature-mined organic photovoltaic device performance database

[Author names and affiliations to be inserted before submission]

Article type: Data Descriptor

Graphical Abstract



Graphical overview of OPV-DB construction, linking harvested literature and tabular data to structured donor-acceptor device records, material identifiers, performance summaries, and technical validation.

Abstract

Organic photovoltaic (OPV) research has generated a large literature of device performance measurements, but these data remain distributed across articles, tables, and heterogeneous reporting formats. We present OPV-DB, a structured literature-mined database containing 40,599 device records from 8,018 source DOIs. Each record links donor and acceptor materials to photovoltaic performance metrics, molecular identifiers, device-stack descriptors, processing fields, and provenance flags where available. The database was produced by a pipeline that combines literature harvesting, full-text and table processing, LLM-assisted extraction, material-name normalization, reference material mapping, and automated validation. PCE is available for 40,341 records (99.4%), all four core

photovoltaic metrics for 38,328 records (94.4%), and both donor and acceptor SMILES for 39,063 records (96.2%). OPV-DB is intended to support benchmarking, data reuse, model development, and comparative studies of OPV materials and devices.

1 Background and Summary

Organic photovoltaics are a flexible and solution-processable solar-cell technology in which device performance depends on coupled choices of donor and acceptor materials, active-layer composition, charge-transport layers, processing conditions, and device architecture. Progress in non-fullerene acceptors and donor polymer design, including Y6-family and related acceptor developments²⁻⁴, has led to rapid improvement in reported power conversion efficiencies. This progress has also produced a data problem: the experimental record is large, diverse, and distributed across thousands of publications. Device data may appear in main-text tables, supplementary tables, paragraphs, and figure captions, with different conventions for naming materials, reporting fill factors, and specifying processing details.

Several OPV datasets have enabled computational studies of molecular properties and photovoltaic performance, including the Harvard Organic Photovoltaic Dataset¹. However, many available resources are limited in scale, time coverage, device-level processing metadata, or direct linkage to the article-level provenance needed for reuse and audit. OPV-DB was constructed to provide a larger device-level resource focused on experimental OPV performance records and associated metadata.

The current OPV-DB release contains 40,599 device records linked to 8,018 source DOIs in the primary device table. The literature harvest contains 19,765 article rows, 19,387 filtered or combined article metadata rows, and 18,874 extraction result rows. The accompanying material reference table contains 4,629 material records. OPV-DB captures 5,162 normalized donor labels, 5,491 normalized acceptor labels, and 13,359 donor-acceptor label pairs. Core photovoltaic coverage is high: PCE is present in 99.4% of device records, Voc in 96.2%, Jsc in 96.0%, FF in 96.1%, and all four metrics in 94.4%. Donor and acceptor SMILES coverage is 97.2% and 98.3%, respectively, with both molecular strings present in 96.2% of records after material-reference mapping.

This Data Descriptor documents the OPV-DB construction workflow, schema, generated data products, reproducible analysis code, technical validation checks, and reuse notes. The dataset is aligned with energy-materials data reuse because it describes characterization and performance data for OPV energy materials while retaining record-level provenance and validation flags so that users can select subsets appropriate for their own tolerance for uncertainty.

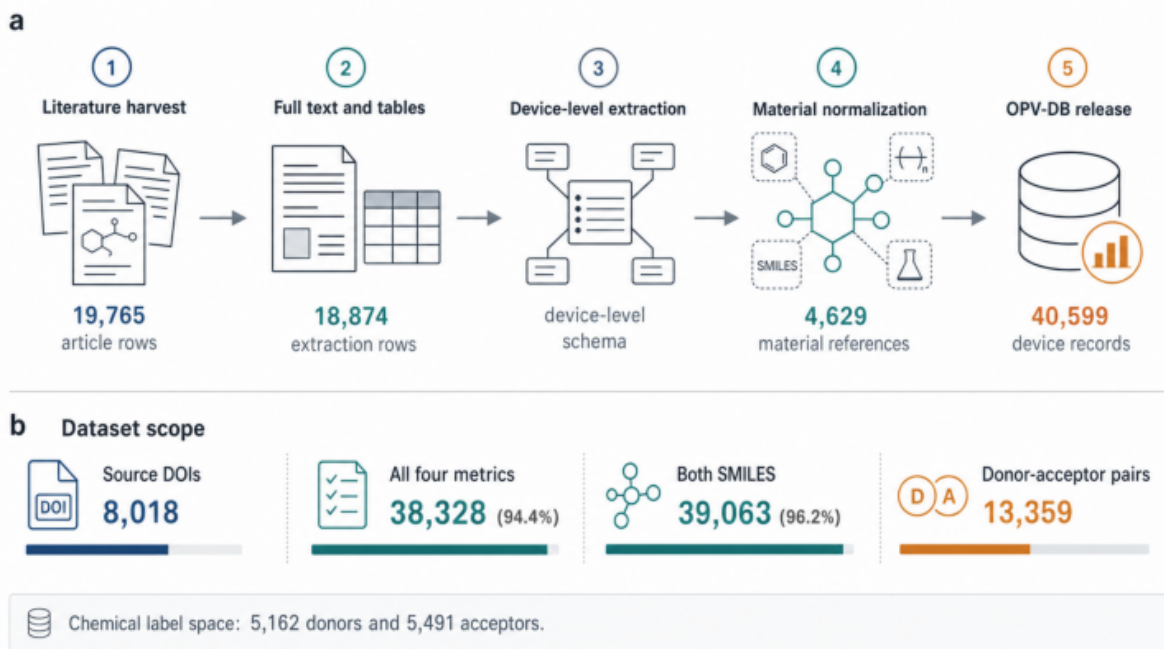


Figure 1: OPV-DB construction workflow and dataset scope. Literature metadata are harvested, full text and tables are processed, device records are extracted into a structured schema, material identifiers are normalized against a reference database, and records are integrated into OPV-DB. Counts reflect the current analysis run.

2 Methods

2.1 Literature harvesting and metadata storage

The OPV literature set was assembled from query-driven harvesting of article metadata related to organic photovoltaics, organic solar cells, donor-acceptor systems, bulk heterojunction devices, photovoltaic performance, and named OPV materials. The harvested metadata include article identifiers, DOI, publisher, title, first author, publication date or year when available, article status fields, and extracted raw article assets. The current article database contains 19,765 rows in the article table and 19,387 rows in both the filtered and combined article metadata tables.

Metadata records were normalized by DOI before linkage to device-level records. Where multiple metadata sources described the same article, combined article records were used to retain a single DOI-level record for downstream joins. Records lacking publication year were retained in the database and excluded only from year-specific summaries.

2.2 Full-text and table processing

Full-text article assets were processed to identify candidate sections and tables containing OPV device performance information. Candidate tables were prioritized when they contained labels or units associated with PCE, Voc, Jsc, FF, donor or acceptor materials, device structures, layer stacks, blend ratios, solvents, additives, active-layer thickness, annealing conditions, or related photovoltaic descriptors.

The extraction workflow stores article-level outputs in an extraction-status database. The current extraction result table contains 18,874 rows. Each row stores DOI, title, processing status, serialized

performance extractions, serialized material information, figure and table counts, processed table counts, extraction time, error messages, and source-file provenance.

2.3 Device-level information extraction

Structured OPV device records were produced from the processed article assets using a schema-focused extraction workflow. The extraction target was a device-level record, not an article-level best value. A single paper may therefore contribute multiple records when it reports multiple donor-acceptor systems, processing conditions, layer stacks, control devices, or optimization variants.

The primary device table stores DOI, donor, acceptor, donor SMILES, acceptor SMILES, Voc, Jsc, FF, PCE, average PCE, best PCE, donor-acceptor ratio, additives, device structure, ETL, HTL, active-layer thickness, solvent, annealing temperature, source-table label, creation and update times, verification flags, donor and acceptor energy levels, canonical material names, canonical solvent/additive/layer names, device type, and quality flags.

Extraction outputs were integrated from article-table extraction, legacy CSV import, and an external curated import. In the current release, 37,147 records are classified as article-table extraction, 2,182 records as legacy CSV import, 1,267 records as external curated import, and 3 records as other article extraction.

2.4 Material-name normalization and molecular identifiers

Material names were normalized to reduce duplicate labels caused by spelling, punctuation, capitalization, and synonym variation. The normalized donor and acceptor labels are stored as `donor_canonical` and `acceptor_canonical`, while the originally extracted labels are retained as `donor` and `acceptor`.

The material reference database contains 4,629 material records with material name, SMILES, material type, chemical class, aliases, frontier orbital energies, and provenance fields where available. Donor and acceptor SMILES in the primary device table were filled by matching extracted material names to this reference resource and by retaining imported molecular strings from curated sources when available. Energy-level fields were likewise mapped from available material-reference entries; users should treat these values as material-reference annotations rather than direct per-article measurements unless source-specific evidence is added in later releases.

2.5 Numerical standardization

Photovoltaic metrics were stored using common OPV units: PCE as percent, Voc as volts, Jsc as mA cm^{-2} , and FF as percent. Active-layer thickness is stored as nanometres when available and annealing temperature as degrees Celsius when available. The analysis workflow coerces numeric columns to numeric values and keeps non-convertible values as missing for summary and validation outputs.

2.6 Acceptors and provenance classes

For analysis figures, acceptors were classified as fullerene or non-fullerene using name-based keyword matching for common fullerene terms such as PC61BM, PC71BM, PCBM, PC60BM, PC70BM, C60, C70, ICBA, bisPC61BM, and fullerene. This classification is intended for descriptive overview figures only. The source dataset retains the original and canonical acceptor names so that users can apply their own stricter chemistry-aware classification.

Provenance classes for manuscript summaries were derived from the source-table field. Values containing **zenodo** were classified as external curated import, values equal to or containing **csv** as legacy CSV import, values beginning with **Table** as article-table extraction, and remaining values as other article extraction.

2.7 Reproducible analysis and figure generation

All summary tables and manuscript figures were generated using a dedicated analysis script. The script reads the current SQLite databases, exports the current device table, writes CSV summary tables, computes coverage and validation statistics, and generates manuscript figures in PNG and PDF formats. The analysis run used pandas, NumPy, matplotlib, seaborn, scipy, and Python's SQLite interface.

The script is intentionally independent of previous report-generation scripts in the repository. It can be re-run after any database update, which avoids manual editing of dataset statistics in the manuscript.

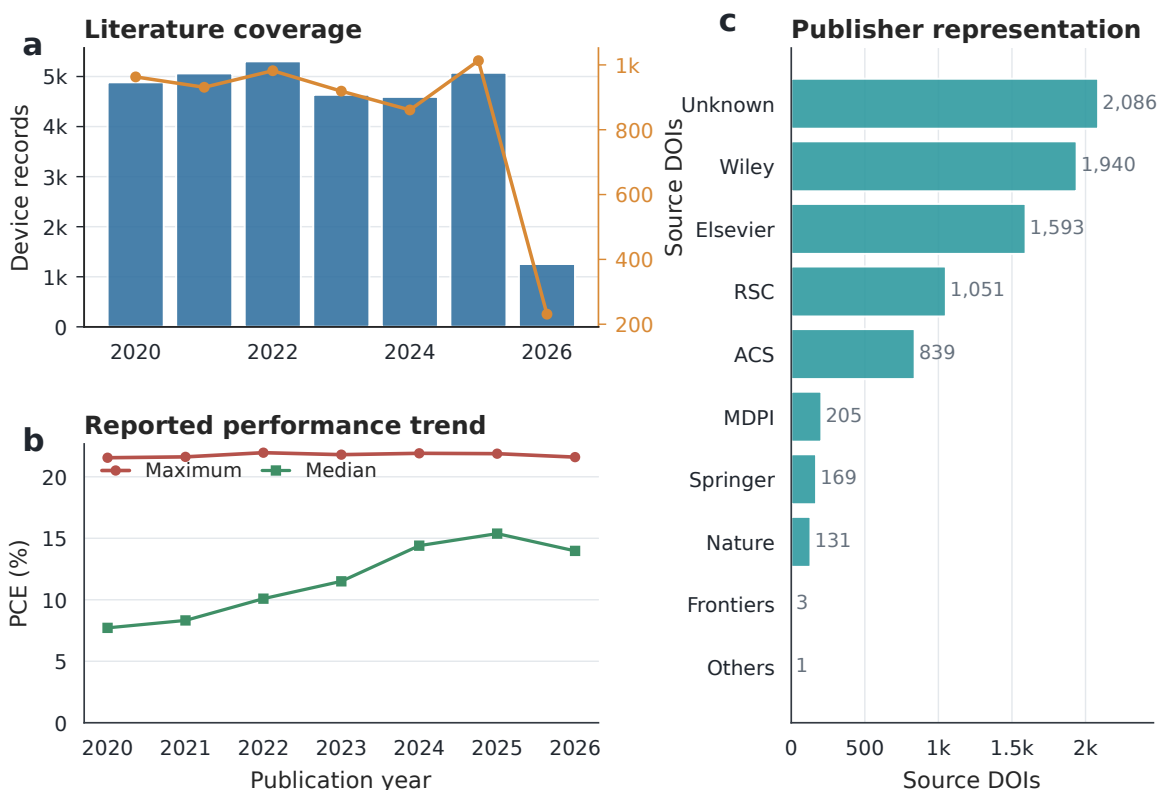


Figure 2: Temporal and publisher coverage. (a) Device records and source DOIs by publication year for records joined to article metadata. (b) Maximum and median PCE by publication year for joined records. (c) Publisher distribution by source DOI. Unknown indicates device records without joined publisher metadata in the current article database.

3 Data Records

The data release is organized around four resource groups: the primary structured device database, article-level metadata and extraction-status resources, a material reference table, and derived release exports. At final submission, the dataset should be deposited in a persistent public repository such as Zenodo, Figshare, or an institutional repository and assigned a DOI. The repository DOI is not yet assigned and is denoted here as `DATA_REPOSITORY_DOI_TBD`.

The primary database contains the `opv_devices` table with 40,599 device records. This is the authoritative structured device database used for the manuscript statistics. Article metadata tables provide DOI, title, publisher, publication year, journal, and article-status metadata used for article-level provenance and temporal summaries. The material reference resource provides material identifiers, SMILES mappings, material types, aliases, and available energy-level annotations. The extraction-status resource preserves article-level extraction outputs and processing status fields.

The release also includes a current device export, a small sample export for inspection, a schema file, generated summary tables, and generated validation tables. These derivative files are intended to document the release and make common reuse tasks easier; they should not replace direct analysis of the database when users need full control over filtering and joins. A file-level inventory is provided in Appendix A.

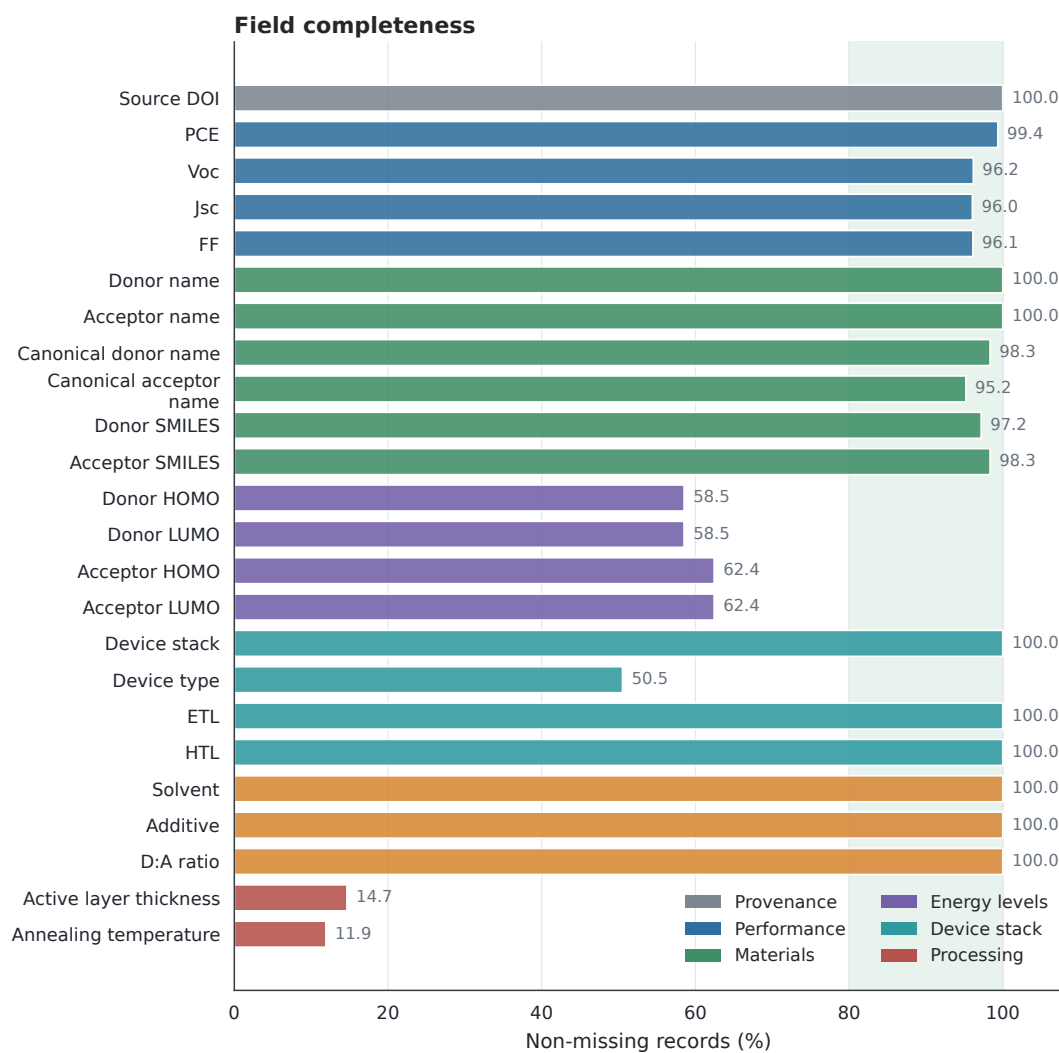


Figure 3: Completeness of manuscript-relevant OPV-DB fields. Field-level non-missing coverage for source identifiers, photovoltaic metrics, material names, molecular identifiers, energy levels, device structure, layer fields, and processing descriptors.

4 Data Overview

OPV-DB contains 40,599 device records. PCE is present in 40,341 records (99.4%), Voc in 39,043 (96.2%), Jsc in 38,994 (96.0%), FF in 39,022 (96.1%), and all four core metrics in 38,328 (94.4%). Donor names are available in 39,928 records (98.3%) and acceptor names in 38,652 (95.2%). Donor SMILES are present in 39,458 records (97.2%), acceptor SMILES in 39,929 (98.3%), and both SMILES in 39,063 records (96.2%) after material-reference mapping.

The PCE distribution has a mean of 9.14%, median of 8.70%, and maximum of 21.96%. Joined publication-year metadata are available for recent records from 2020 through 2026, with median PCE increasing from 7.72% in 2020 to 15.38% in 2025 before the partial 2026 data.

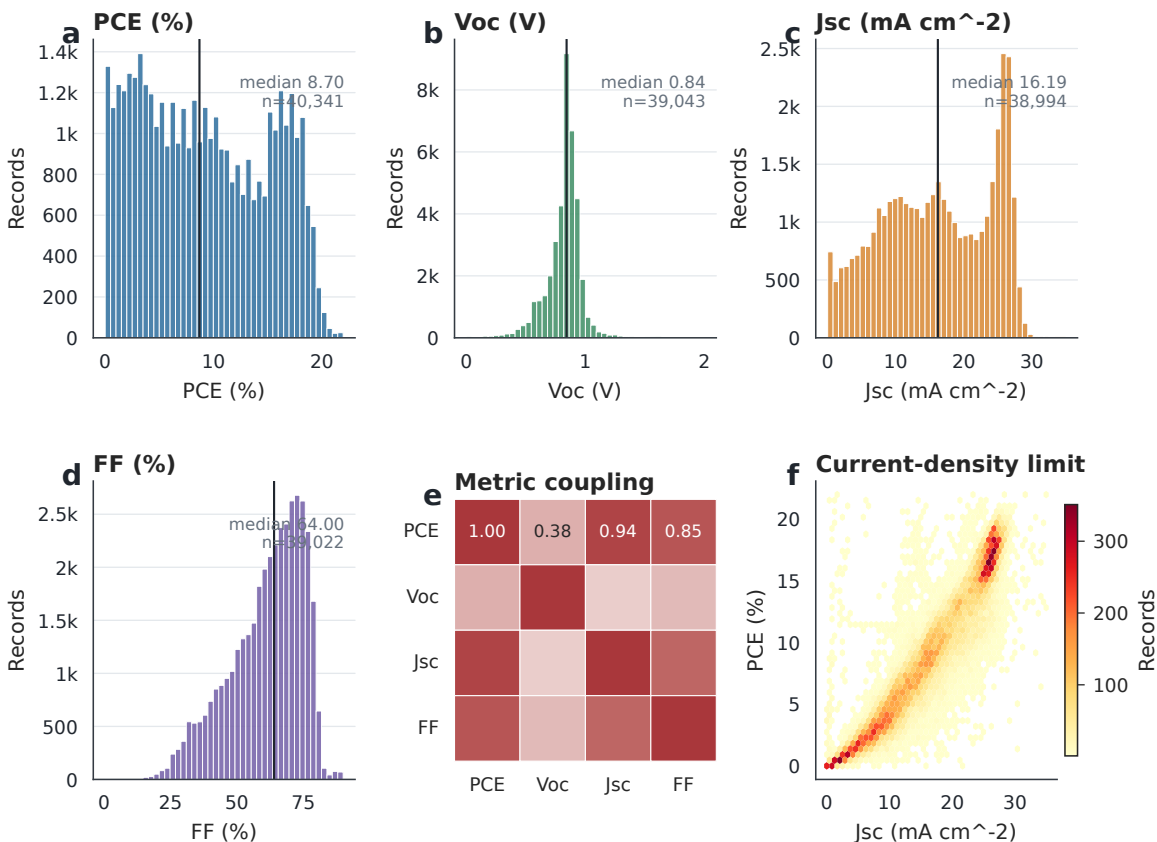


Figure 4: Distribution and coupling of photovoltaic performance metrics. Histograms summarize PCE, Voc, Jsc, and FF. The correlation heatmap and density plot show relationships among the four core metrics for records passing broad physical filters.

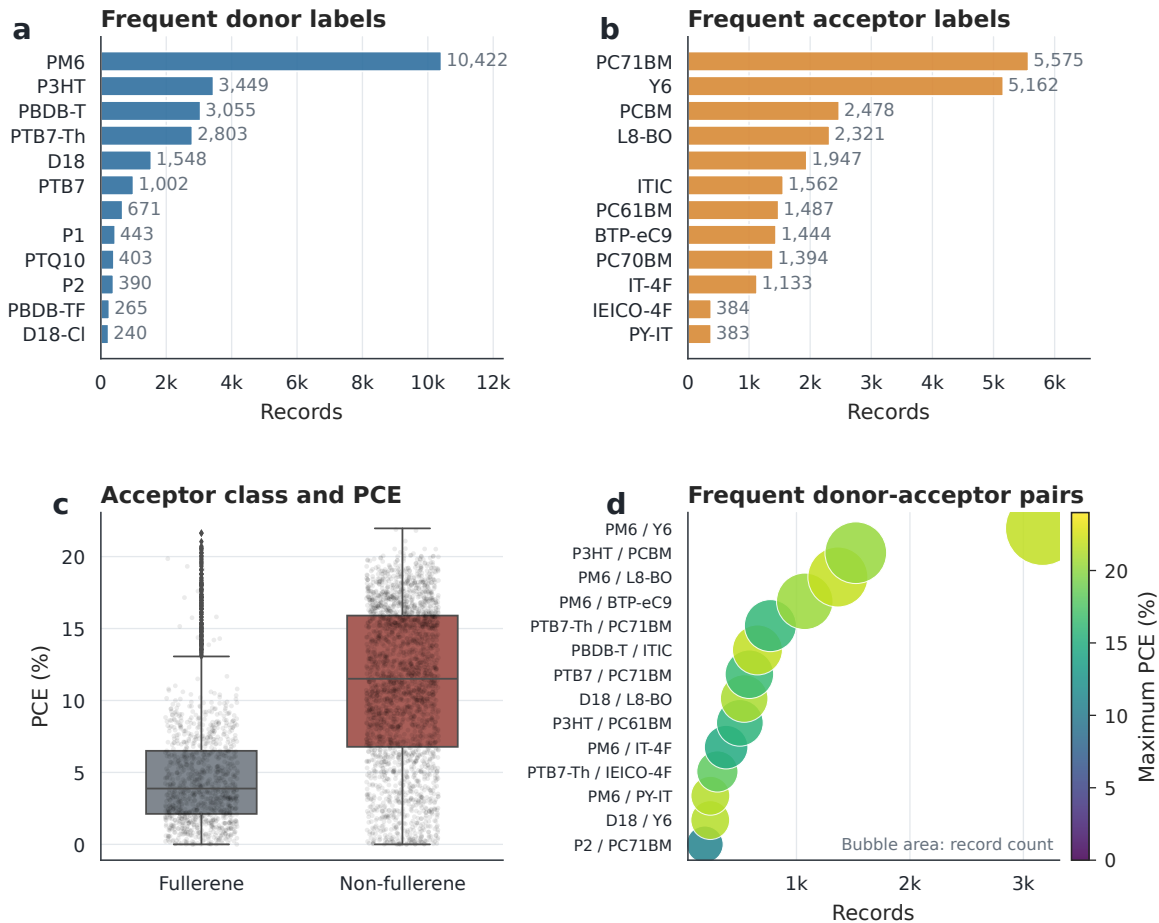


Figure 5: OPV material and donor-acceptor landscape. (a,b) Most frequent donor and acceptor labels. (c) PCE distributions for fullerene and non-fullerene acceptor records using name-based classification. (d) Frequent donor-acceptor pairs, with bubble size proportional to maximum PCE.

5 Technical Validation

5.1 Schema and missingness validation

The analysis workflow computes field-level coverage for key variables and writes the results to a release validation table. Required provenance fields, including DOI and source-table label, are present in all 40,599 device records. The core performance and material-identifier fields have high coverage. Processing columns such as solvent, additive, and donor-acceptor ratio are retained as reported or normalized label fields, but users should filter placeholder values for analyses that require explicitly reported process settings. Quantitative processing descriptors remain less frequently available: active-layer thickness coverage is 14.7%, and annealing temperature coverage is 11.9%.

5.2 Physical range validation

The validation workflow applies broad physical range checks to numeric fields. All available PCE values fall within 0–30%, all available Voc values within 0–2.5 V, all available Jsc values within 0–50 mA cm⁻², and all available FF values within 0–100%. Active-layer thickness values pass the broad 0–1000 nm range in 99.8% of checked records, and annealing temperatures pass the broad 0–300 °C range in 99.7% of checked records. Records outside these broad processing ranges are retained in the dataset so that users can inspect and decide whether they represent extraction errors, unusual experiments, or unit-reporting differences.

5.3 Photovoltaic equation consistency

For 38,328 records with PCE, Voc, Jsc, and FF, the workflow recalculates PCE using

$$\text{PCE} = \frac{V_{\text{oc}} \times J_{\text{sc}} \times FF}{100}.$$

The recomputed value is within 5% relative error of the reported PCE for 35,892 records (93.6%) and within 10% relative error for 37,036 records (96.6%). Larger deviations can arise from rounded values, unit inconsistencies, copied table errors, extraction errors, or records where reported average and best values were mixed in the source text. These deviations are preserved in the release validation records for audit and downstream filtering.

5.4 Verification flags

The `verified` field records current validation and review status. The present release contains 21,744 records flagged `AI_ACCEPT`, 6,601 flagged `AI_SPOT_CHECK`, 812 flagged `AI_MANUAL`, 2,064 flagged `YES`, and 9,378 flagged `NO`. These flags should be interpreted as provenance and review-state indicators rather than a replacement for a complete human audit. Before final journal submission, a manually adjudicated validation sample should be completed and reported if the journal or reviewers require human accuracy estimates for LLM-extracted fields.

5.5 Known limitations

Energy-level annotations are mapped from the material reference database when available and are not guaranteed to be the exact energy-level values measured in each source article. Some material names, especially short internal labels such as P1, P2, and P3, can be ambiguous across papers. The dataset keeps both raw and canonical material labels so that users can select conservative filters, for example by requiring both donor and acceptor SMILES. Temporal summaries are limited to

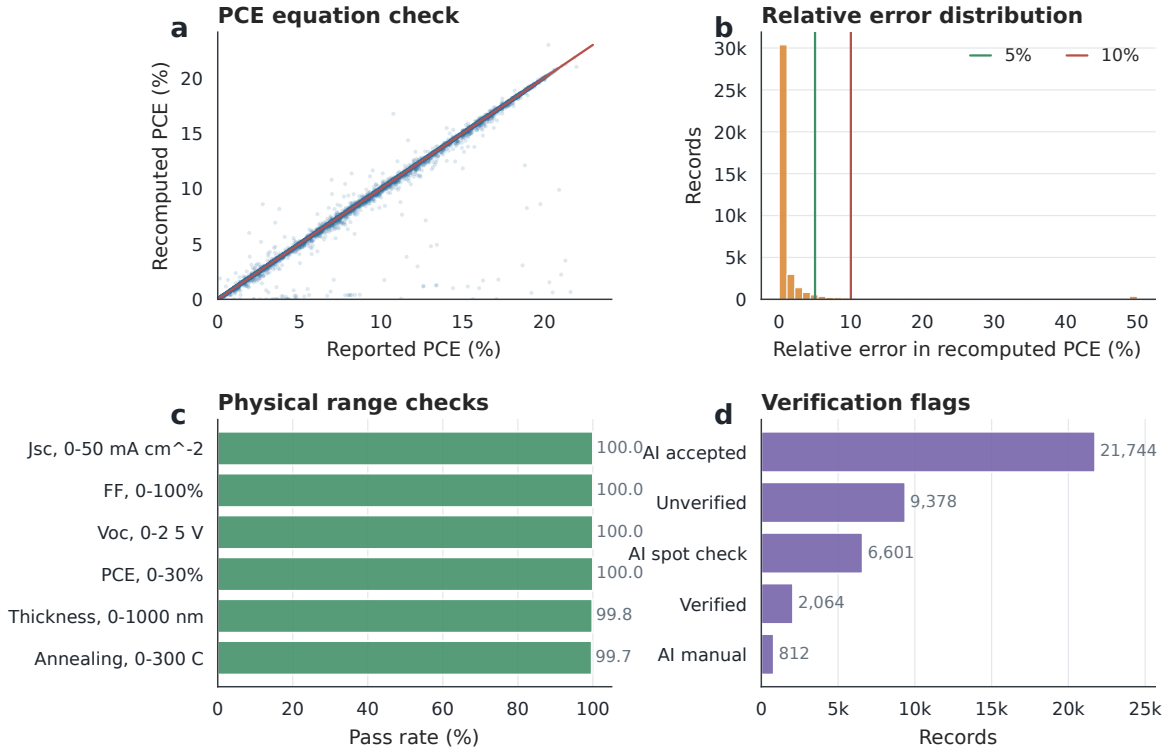


Figure 6: Technical validation summaries. (a) Reported PCE compared with PCE recomputed from Voc, Jsc, and FF. (b) Relative error distribution for the PCE consistency check. (c) Broad physical range checks for key numeric fields. (d) Verification and AI review flags in the current database.

records that can be joined to publication-year metadata; records without year metadata remain in the device database but are excluded from year-specific plots.

6 Usage Notes

Users interested in device-performance modelling should start with records that contain PCE, Voc, Jsc, FF, donor, acceptor, and DOI. Users interested in molecular modelling should additionally require donor and acceptor SMILES. Users interested in process-property relationships should inspect coverage and missingness for solvent, additive, donor-acceptor ratio, thickness, annealing, ETL, HTL, and device structure before fitting models.

The recommended reproducibility workflow is to load the current device export or query the primary database directly, normalize DOI strings to lowercase, and filter by `verified`, `data_quality_flag`, `source_table`, and completeness requirements appropriate to the analysis. For strict studies, users can require all four core performance metrics, both SMILES fields, and PCE equation consistency within a chosen threshold. For broader exploratory studies, users may retain records with missing processing fields because these fields are not uniformly reported in the OPV literature.

The generated CSV tables are intended to document the current release and should not replace direct analysis of the database. Users should cite the repository DOI once assigned and should report the release date or database hash used for their analyses.

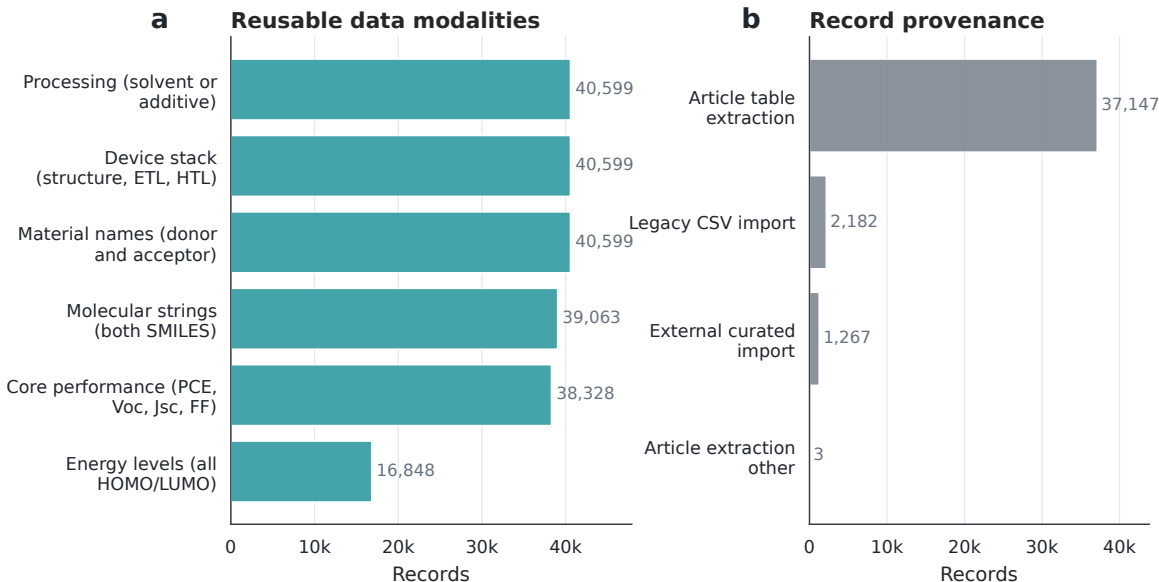


Figure 7: Data modalities and provenance. (a) Number of records supporting common reuse tasks, including core performance modelling, material-name analysis, SMILES-based modelling, energy-level analysis, device-stack analysis, processing analysis, and other reuse modes. (b) Source groups derived from record provenance.

7 Data Availability

The OPV-DB dataset described in this Data Descriptor will be deposited in a persistent public repository before journal submission. The final data repository DOI is DATA_REPOSITORY_DOI_TBD. If publisher full-text content cannot be redistributed, the public repository should include derived metadata, extracted structured records, extraction status, article DOI lists, and code sufficient to reproduce extraction for users with lawful access to the source articles.

8 Code Availability

The analysis code used to generate the manuscript assets will be archived with the data release or tagged in a public source-code repository. The final code repository URL or DOI is CODE_REPOSITORY_URL_OR_DOI_TBD.

9 Acknowledgements

[Acknowledgements to be added.]

10 Funding

[Funding information to be added.]

11 Author Contributions

[Author contribution statement to be added. Suggested structure: Conceptualization: X.X.; Data curation: X.X.; Software: X.X.; Validation: X.X.; Formal analysis: X.X.; Writing original draft: X.X.; Writing review and editing: all authors; Supervision: X.X.; Funding acquisition: X.X.]

12 Competing Interests

The authors declare no competing interests. [Confirm before submission.]

13 References

1. Lopez, S. A., Pyzer-Knapp, E. O., Simm, G. N., Lutzow, T., Li, K., Seress, L. R., Hachmann, J. & Aspuru-Guzik, A. The Harvard organic photovoltaic dataset. *Scientific Data* **3**, 160086 (2016). <https://doi.org/10.1038/sdata.2016.86>
2. Yuan, J. et al. Single-junction organic solar cell with over 15% efficiency using fused-ring acceptor with electron-deficient core. *Joule* **3**, 1140–1151 (2019). <https://doi.org/10.1016/j.joule.2019.01.004>
3. Yu, Z.-P. et al. Simple non-fused electron acceptors for efficient and stable organic solar cells. *Nature Communications* **10**, 2152 (2019). <https://doi.org/10.1038/s41467-019-10098-z>

4. Cui, Y. et al. Over 16% efficiency organic photovoltaic cells enabled by a chlorinated acceptor with increased open-circuit voltages. *Nature Communications* **10**, 2515 (2019).
<https://doi.org/10.1038/s41467-019-10351-5>

A Release File Inventory

This appendix lists the release files that should accompany the public data deposition. These names describe the archived data package rather than a local author workspace.

Resource group	File or table	Purpose
Primary device database	<code>opv_performance.db</code> ; table <code>opv_devices</code>	Authoritative device-level OPV performance records.
Article metadata	<code>OrganicPV-2025-Jan_articles.db</code> or redistributable metadata export	Source DOI, title, publisher, year, journal, and article-status metadata.
Material references	<code>materials.db</code>	Material names, aliases, SMILES, material classes, and mapped energy-level annotations.
Extraction status	<code>extraction_results.db</code> or redistributable extraction-status export	Article-level extraction outputs, status fields, and processing diagnostics.
Device export	<code>opv_devices_current_export.csv</code>	Flat CSV export of manuscript-relevant device fields.
Sample export	<code>opv_devices_sample250.csv</code>	Small sample for inspection and documentation.
Schema	<code>opv_devices_schema.csv</code>	Exported columns and inferred data types.
Summary tables	<code>table1_dataset_summary.csv</code> through <code>table9_source_distribution.csv</code>	Dataset counts, coverage, descriptive statistics, material frequencies, yearly trends, publisher summaries, and source distribution.
Validation tables	<code>table10_pce_consistency_records.csv</code> through <code>table15_extraction_status.csv</code>	PCE consistency records, validation checks, verification flags, material-reference summaries, article-harvest summaries, and extraction-status summaries.
Figures	Figure files in PNG and PDF formats	Manuscript figures and reusable graphical summaries.

B Data Dictionary

The following dictionary describes the primary fields exported for the device-level records.

id (integer) Unique device-record identifier.

doi (string) Source publication DOI.

donor (string) Donor material label extracted from the source.

acceptor (string) Acceptor material label extracted from the source.

donor_canonical (string) Normalized donor label.

acceptor_canonical (string) Normalized acceptor label.

donor_smiles (string) Donor molecular SMILES when available.

acceptor_smiles (**string**) Acceptor molecular SMILES when available.

voc (**float**) Open-circuit voltage in V.

jsc (**float**) Short-circuit current density in mA cm^{-2} .

ff (**float**) Fill factor in percent.

pce (**float**) Power conversion efficiency in percent.

pce_avg (**float**) Average PCE if reported.

pce_best (**float**) Best PCE if reported.

d_a_ratio (**string**) Donor:acceptor blend ratio as reported or normalized.

additive (**string**) Additive information as reported.

additive_canonical (**string**) Normalized additive label.

additive_ratio (**string/float**) Additive amount when available.

device_structure (**string**) Reported device stack or structure.

device_type (**string**) Conventional, inverted, or other device type when inferred.

etl (**string**) Electron transport layer as reported.

etl_canonical (**string**) Normalized ETL label.

htl (**string**) Hole transport layer as reported.

htl_canonical (**string**) Normalized HTL label.

active_layer_thickness (**float**) Active-layer thickness in nm when available.

solvent (**string**) Processing solvent as reported.

solvent_canonical (**string**) Normalized solvent label.

annealing_temp (**float**) Annealing temperature in degrees Celsius when available.

homo_d (**float**) Donor HOMO energy in eV from material reference mapping.

lumo_d (**float**) Donor LUMO energy in eV from material reference mapping.

eg_d (**float**) Donor optical or electronic gap in eV when available.

homo_a (**float**) Acceptor HOMO energy in eV from material reference mapping.

lumo_a (**float**) Acceptor LUMO energy in eV from material reference mapping.

eg_a (**float**) Acceptor optical or electronic gap in eV when available.

source_table (**string**) Table, import, or source label from which the record was derived.

data_source_type (**string**) Broad source type flag when available.

data_quality_flag (**string**) Data quality annotation when available.

verified (**string**) Verification or AI review flag.

is_baseline_ref (**string/integer**) Baseline reference flag when available.

C Recommended Filters and Units

Core units are PCE in percent, Voc in V, Jsc in mA cm^{-2} , FF in percent, active-layer thickness in nm, and annealing temperature in degrees Celsius. For core device analysis, users should require `pce`, `voc`, `jsc`, `ff`, `donor`, and `acceptor`. Molecular modelling should additionally require `donor_smiles` and `acceptor_smiles`. Strict performance-consistency studies should require PCE equation relative error below a selected threshold. Provenance-sensitive analyses should filter by `verified`, `source_table`, or `data_quality_flag`.